

Project Report

STEREO VISION DEPTH ESTIMATOR -

A RANSAC-Based Epipolar Geometry Pipeline for Dense Depth Estimation

Course: Computer Vision (CSE3010)

Prepared by : ARJUN TIWARI

Registration No. : 24BAI10141

B.Tech CSE (AI/ML), VIT Bhopal University

1. Introduction

Depth estimation is the recovery of scene distance from image observations. In binocular stereo vision, two horizontally separated cameras observe the same scene. A 3D point appears at different image coordinates in the two views; this displacement, called disparity, provides the information required for depth recovery.

Stereo vision: A technique that estimates scene depth by comparing corresponding observations from two viewpoints.

Disparity: The difference in image position between corresponding points in the left and right views.

Depth map: An image whose valid pixels contain estimated distances from the camera.

Epipolar geometry: The geometric relationship between two camera views that constrains possible point correspondences.

The pipeline combines calibration, SIFT feature matching, custom RANSAC estimation of the Fundamental matrix, pose recovery, stereo rectification, SGBM disparity estimation, and metric depth conversion.

2. Problem Statement

A single 2D image does not directly encode the distance of every visible point. The system therefore aims to recover dense depth from a stereo image pair while handling incorrect matches and geometric errors.

- Find reliable correspondences.
- Reject outliers robustly.
- Estimate relative camera geometry.
- Rectify the stereo pair.
- Compute dense disparity.
- Convert disparity to depth and evaluate the result.

3. Objectives

1. Implement a complete two-view stereo pipeline.
2. Demonstrate the normalized eight-point algorithm with RANSAC.
3. Use SIFT correspondences and epipolar constraints.
4. Generate dense disparity using SGBM.
5. Recover depth using $Z = fB/d$.
6. Validate results with automated tests and metrics.

4. Functional Requirements

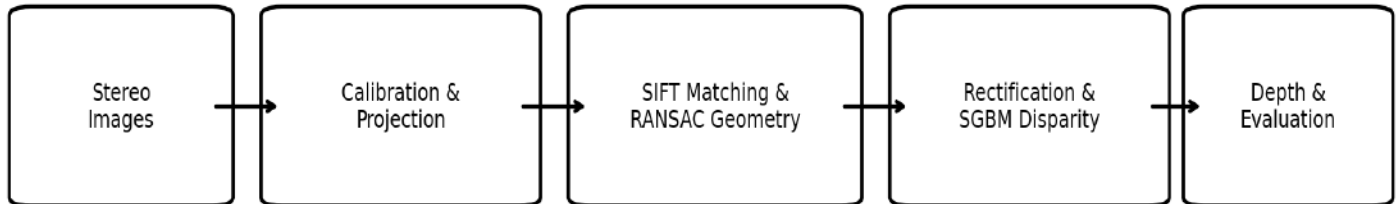
- Module 1 — Calibration & Projection: maintain K , $[R|t]$, projection matrices, and undistort images.
- Module 2 — Feature Matching & Epipolar Geometry: SIFT matching, custom RANSAC, Fundamental/Essential matrices, and relative pose.
- Module 3 — Disparity & Depth: rectification, SGBM disparity, and metric depth.
- Input/Output: two image paths in; depth map, diagnostics, and JSON metrics out.
- Workflow: load → match → geometry → rectify → disparity → depth → evaluate.

5. Non-Functional Requirements

- Performance: approximately under four seconds for the stated 1282×1110 pair on CPU.
- Reliability: floor the RANSAC inlier-ratio estimate before adaptive trial calculation.
- Error handling: explicit exceptions for missing/invalid/insufficient inputs.
- Logging: progress, match counts, inlier ratios, and disparity coverage.
- Maintainability: separate modules with independent tests.
- Resource efficiency: avoid redundant disparity computation.

6. System Architecture

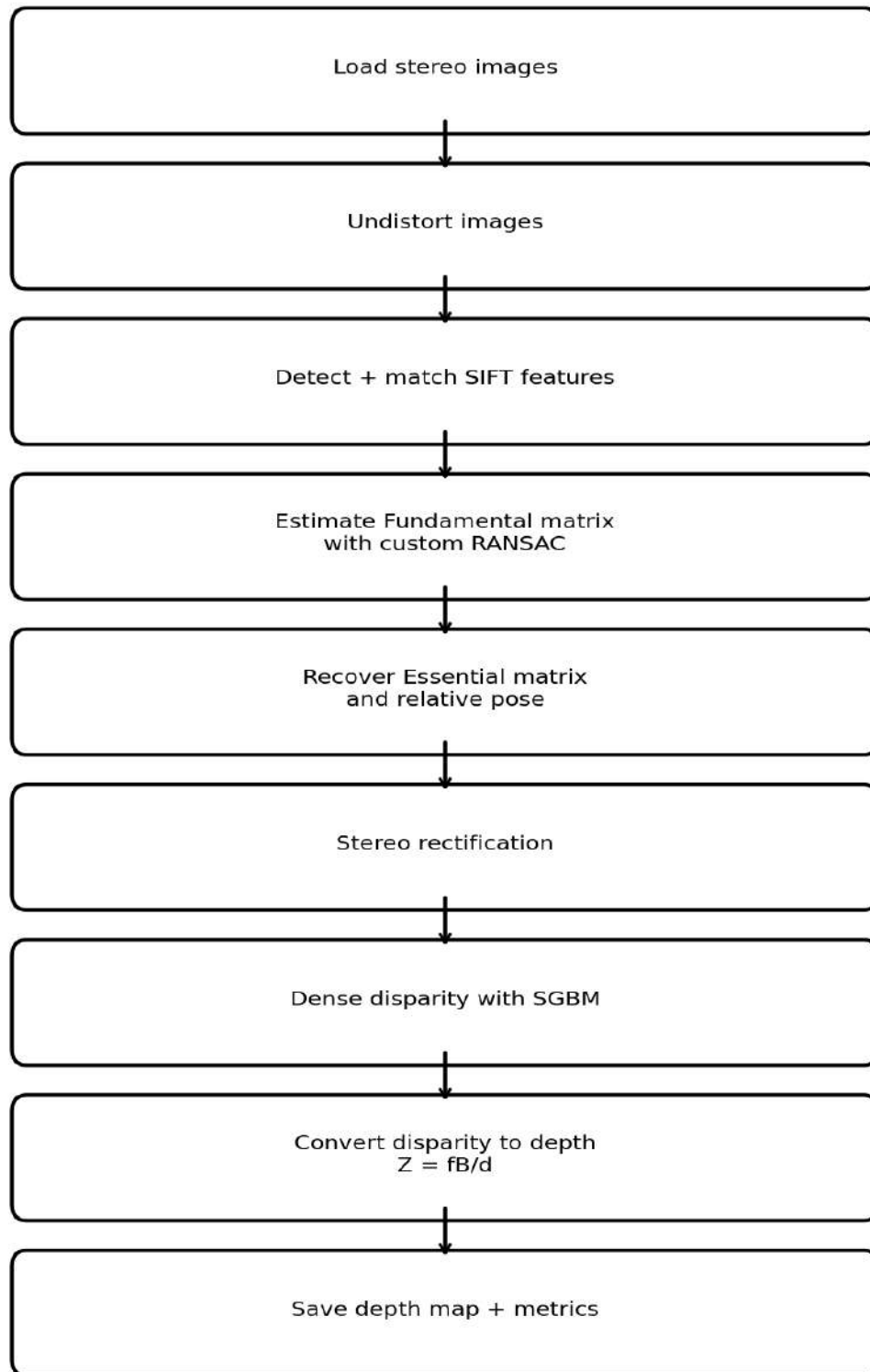
Figure 2. System Architecture



1. **Stereo Image Input:** The system takes a pair of horizontally offset stereo images as the primary input for depth estimation.
2. **Calibration & Projection:** Camera parameters and projection models are used to prepare the images and establish the geometric relationship between the camera views.
3. **Feature Matching & RANSAC Geometry:** SIFT features are detected and matched between the images, while RANSAC is used to estimate reliable epipolar geometry and reject incorrect matches.
4. **Rectification & Disparity Estimation:** The stereo images are rectified so corresponding points align along the same scanlines, after which SGBM is used to generate a dense disparity map.
5. **Depth & Evaluation:** The disparity map is converted into metric depth, and the resulting depth map is evaluated using quantitative metrics and visual outputs.

7. Process Workflow

Figure 1. End-to-End Stereo Depth Workflow



8. Core Concepts and Definitions

Camera intrinsic matrix (K): Contains internal camera parameters such as focal lengths and principal point.

Extrinsic parameters: Describe camera rotation and translation relative to a reference frame.

Projection matrix: Maps a 3D point in homogeneous coordinates to image coordinates.

SIFT: A scale-invariant local feature detector and descriptor used to establish image correspondences.

Fundamental matrix (F): A 3×3 rank-2 matrix expressing the epipolar constraint between two camera views.

Essential matrix (E): Encodes relative rotation and translation for calibrated views; here $E = K^*T.F.K$.

RANSAC: A robust model estimation method that finds a model supported by a consistent subset of observations.

Sampson distance: A first-order geometric error for measuring correspondence consistency with an epipolar model.

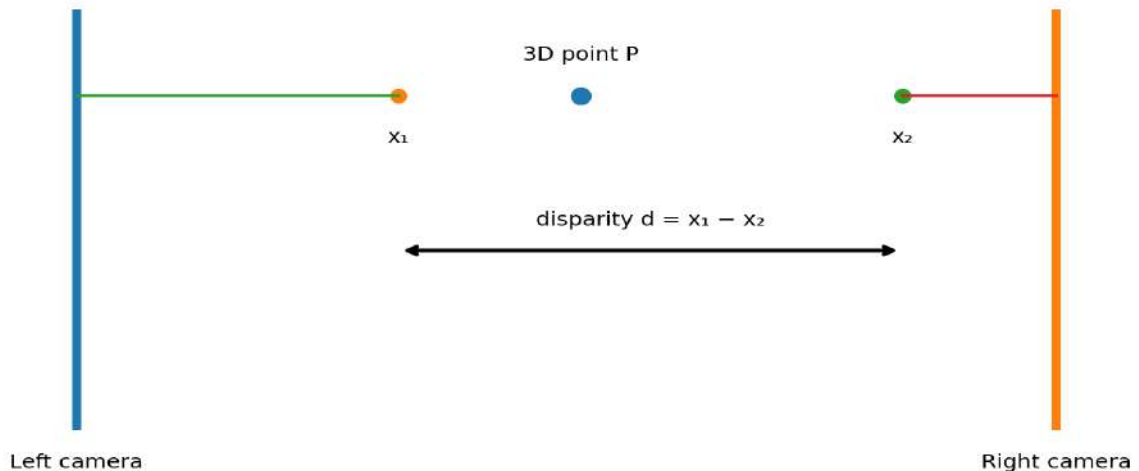
Stereo rectification: Transforms stereo images so corresponding points lie on corresponding horizontal scanlines.

SGBM: Semi-Global Block Matching, a stereo method that combines matching evidence with smoothness constraints.

Triangulation: Recovers a 3D point from corresponding image observations and camera geometry.

9. Stereo Geometry Visual Aid

Figure 3. Stereo Disparity Concept



In a rectified stereo setup, the same 3D point appears at different horizontal positions in the left and right images.

These corresponding image positions are represented by x_1 and x_2 , respectively.

The difference between these positions is called **disparity**, which is given by $d = x_1 - x_2$.

Disparity provides an important cue for determining how far an object is from the cameras.

A larger disparity tells that the object is closer, whereas a smaller disparity indicates greater depth.

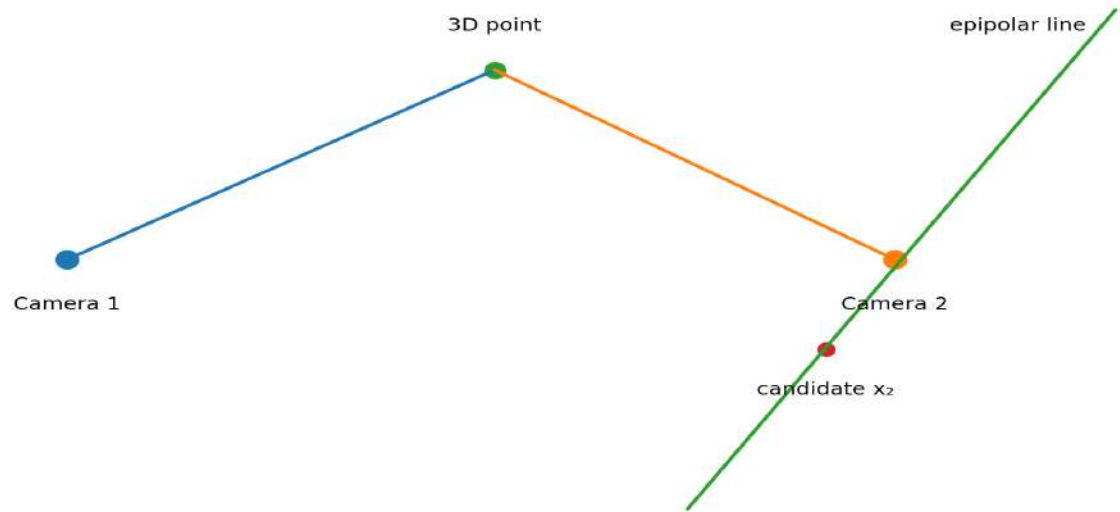
The system converts disparity into metric depth using the formula $Z = (f * B) / d$, where f is the focal length and B is the camera baseline.

* For a rectified stereo setup disparity and depth always follow an inverse relationship as :
larger disparity corresponds to smaller depth.

In which the implementation uses $Z = (f \times B) / d$.

10. Epipolar Geometry Visual Aid

Figure 4. Epipolar Geometry



Epipolar geometry describes the geometric relationship between two camera views observing the same 3D point.

When a point is observed in one image, its corresponding point in the second image is constrained to lie along a specific **epipolar line**.

This constraint significantly reduces the search area for finding corresponding points between the two images.

A point in one image induces an epipolar line in the other image. Correct correspondences should satisfy the associated geometric constraint.

In the diagram, the 3D point is projected onto both cameras, establishing the corresponding image observations.

RANSAC helps identify reliable correspondences and reject incorrect matches before estimating the Fundamental matrix.

The resulting epipolar geometry is then used to support stereo rectification and accurate disparity-based depth estimation.

11. Algorithm Design

11.1 Feature Detection and Matching

7. Detect SIFT keypoints.
8. Compute descriptors.
9. Match descriptors between views.
10. Apply the ratio test.
11. Send surviving point pairs to geometric estimation.

11.2 Fundamental Matrix with RANSAC

12. Sample eight correspondences.
13. Normalize coordinates.
14. Solve the linear system with SVD.
15. Enforce rank-2 by zeroing the smallest singular value.
16. Score correspondences using Sampson distance.
17. Keep the strongest inlier set.
18. Adapt the remaining trial count using the RANSAC formula.

$$N = \log(1 - p) / \log(1 - w^{**}s)$$

11.3 Pose Recovery and Rectification

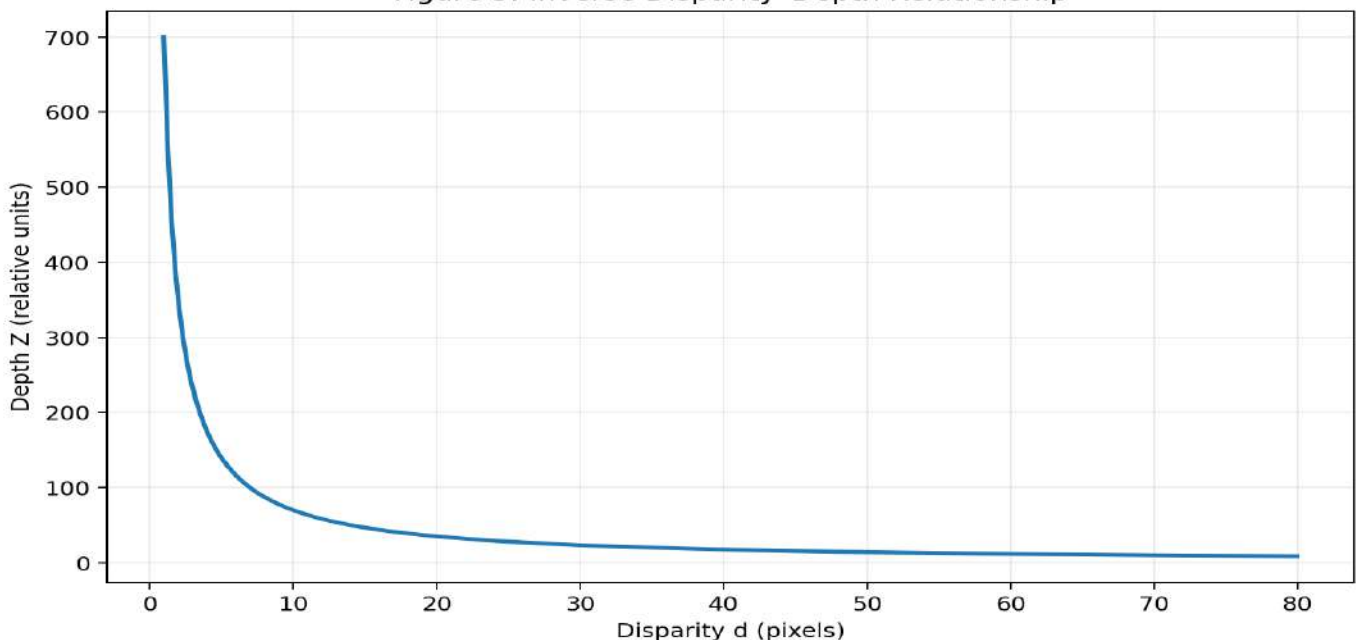
The Essential matrix is obtained from $E = K^T F K$. Relative rotation and translation are recovered, then the stereo pair is rectified so matching becomes primarily a row-wise search.

11.4 Dense Disparity and Depth

SGBM estimates dense disparity over the rectified images. Valid disparity is converted to depth using:

$$Z = (f * B) / d$$

Figure 5. Inverse Disparity-Depth Relationship



12. Component Design

Component	Responsibility	Output
config.py	Parameters and camera settings	Configuration
calibration.py	Undistortion and camera geometry	Calibration data
epipolar.py	SIFT, matching, RANSAC, F/E, pose	Geometry
depth.py	Rectification, SGBM, depth conversion	Disparity + depth
evaluate.py	Validation and metrics	JSON metrics
visualize.py	Plots and overlays	Visual diagnostics
utils.py	Shared utilities and I/O	Helpers
main.py	Pipeline orchestration	End-to-end run

13. Design Decisions and Rationale

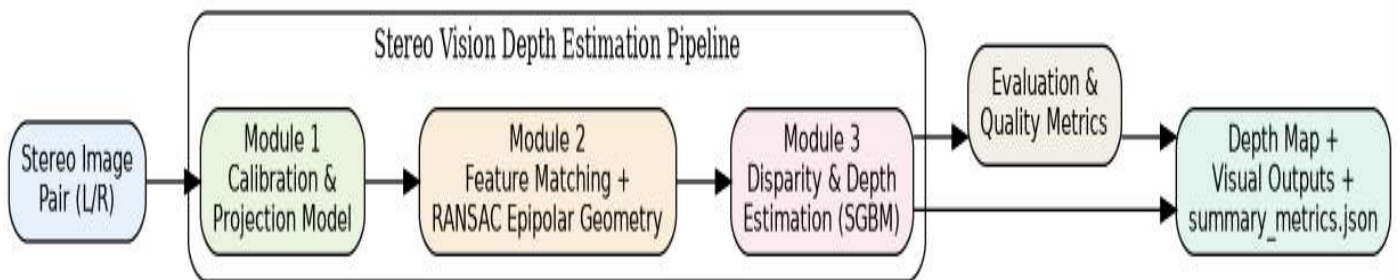
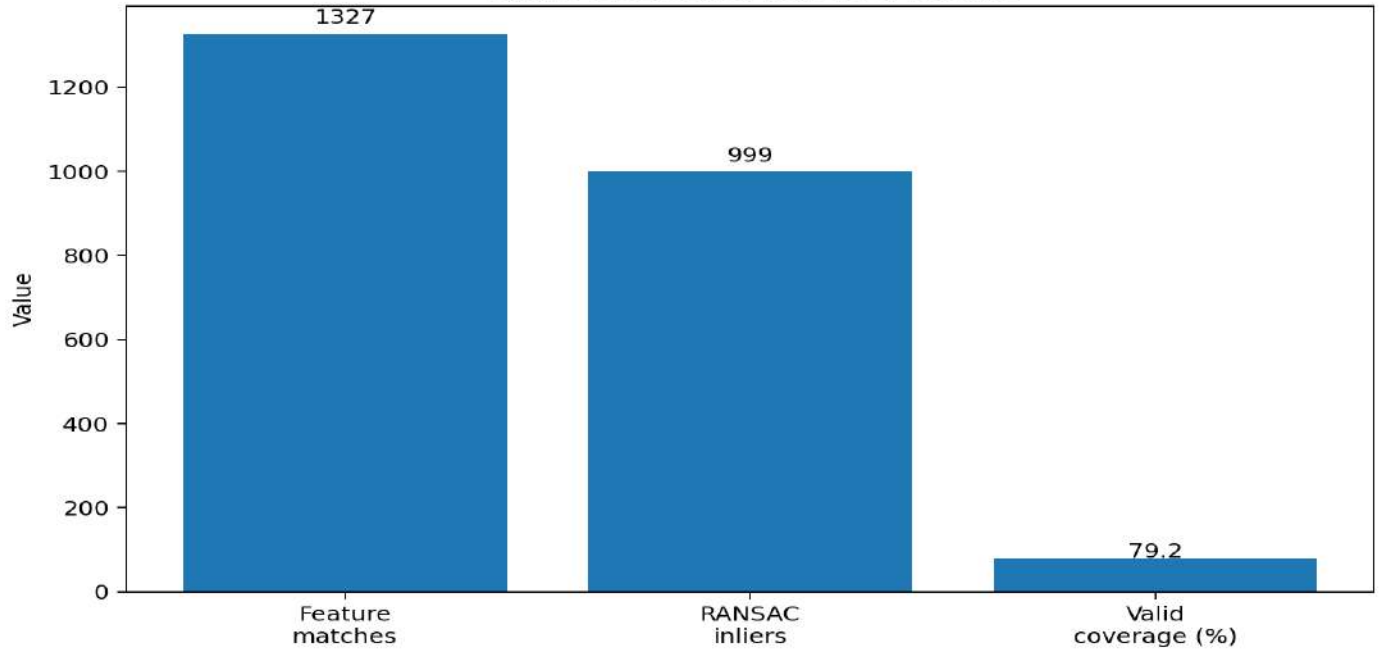
- Custom RANSAC exposes the normalized eight-point and Sampson-distance steps.
- SIFT was selected for stable correspondences on the textured benchmark scene.
- SGBM was selected for smoother and more complete disparity than simple block matching.
- The modular layout supports independent testing and maintenance.
- Simulated intrinsics are used when physical checkerboard calibration is unavailable.

14. Dataset Description

The reference implementation was validated on the Middlebury 'Aloe' stereo pair at 1282×1110 resolution. It is a textured potted-plant scene suitable for demonstrating disparity and depth separation. The pipeline accepts other horizontally offset stereo pairs through left/right image inputs.

15. Visual Results

Figure 6. Selected Reported Results





SIFT feature matches before RANSAC filtering



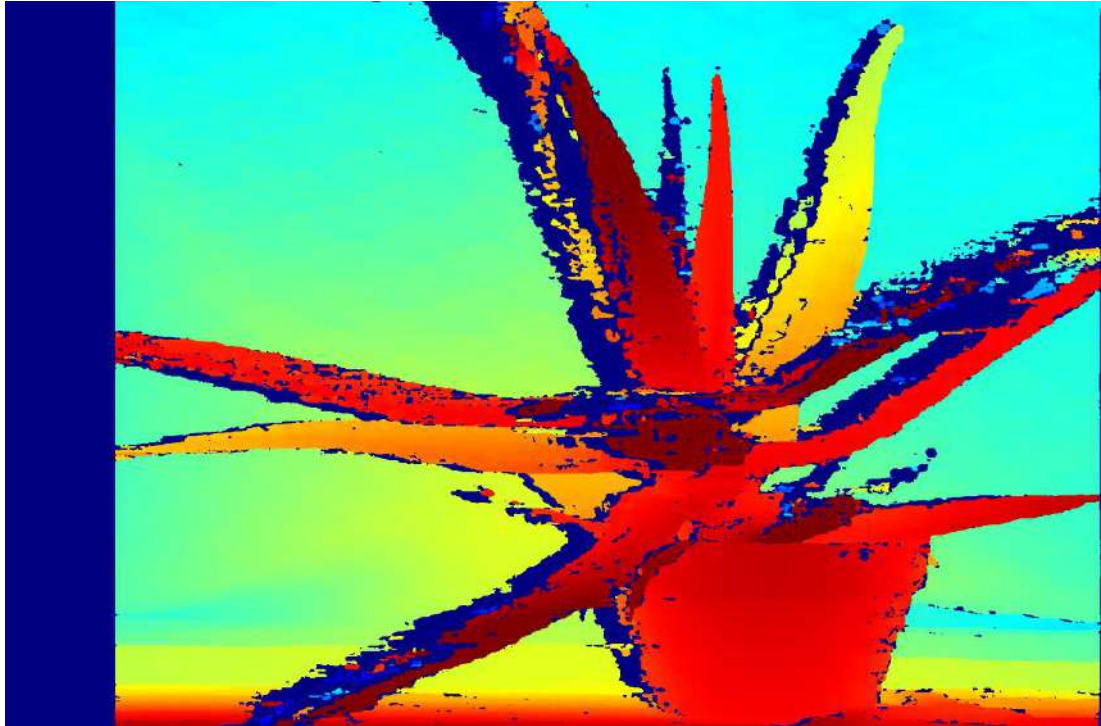
RANSAC inlier matches (outliers rejected)



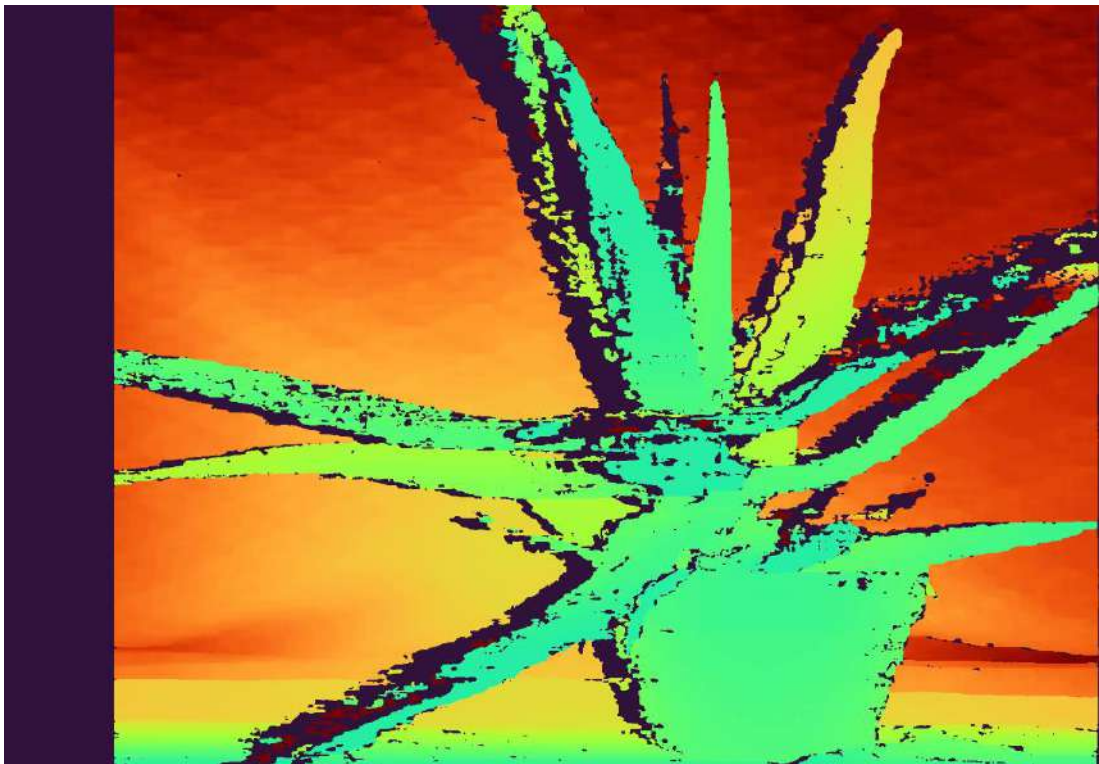
Epipolar line overlay on the right image



Stereo-rectified pair



Embedded visual output from the reference report



Embedded visual output from the reference report

16. Quantitative Results

Metric	Reported value
Feature matches after ratio test	1327
RANSAC inliers	999
Reported RANSAC inlier ratio	99.8%
Fundamental matrix rank	2
Mean epipolar residual	0.1352
Disparity valid coverage	79.2%
Depth valid pixels	79.2%
Mean estimated depth	3.466 m
Pipeline runtime	3.64 s

The values above reproduce the measurements stated in the supplied reference report.

17. Testing Approach

The reference report describes nine automated pytest tests covering calibration shapes, matching validity, Fundamental-matrix rank, RANSAC inlier ratio, epipolar residual, disparity coverage, disparity-depth behavior, NaN handling, and missing-file errors. It reports that all nine tests pass.

18. Challenges Faced

- Adaptive RANSAC trial scheduling could collapse after an unlucky early sample; a floor was added to the inlier-ratio estimate.
- SGBM parameters require balancing coverage and noisy matches.
- Invalid disparity regions must be handled consistently through the depth map and visualization.

19. Learnings and Key Takeaways

- Rank-2 enforcement and Sampson distance improve the robustness of Fundamental-matrix estimation.
- Numerical-stability bugs can silently degrade a model, making automated tests important.
- Rectification quality strongly affects downstream disparity quality.
- Modular design makes the pipeline easier to understand, test, and extend.

20. Future Enhancements

- Use real checkerboard-based camera calibration.
- Add a Streamlit interface for parameter tuning.
- Extend toward multi-view Structure-from-Motion and sparse 3D visualization.
- Investigate GPU acceleration or learned stereo matching for real-time video depth.

21. References

- R. Hartley and A. Zisserman, Multiple View Geometry in Computer Vision, Cambridge University Press.
- M. A. Fischler and R. C. Bolles, “Random Sample Consensus: A Paradigm for Model Fitting with Applications to Image Analysis and Automated Cartography,” Communications of the ACM, 1981.
- D. Scharstein and R. Szeliski, Middlebury Stereo Vision Datasets.
- OpenCV documentation — Camera Calibration and 3D Reconstruction module.